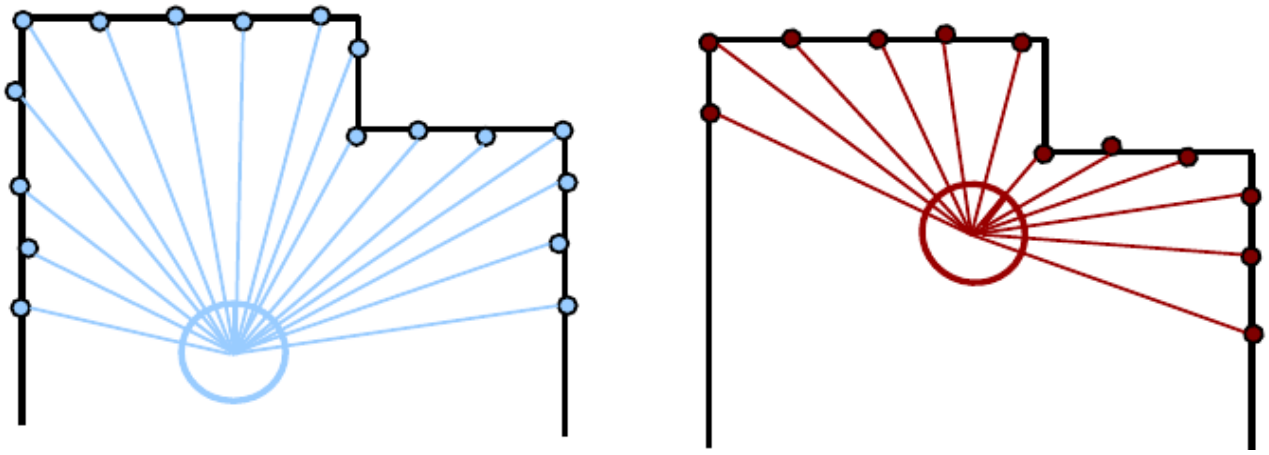
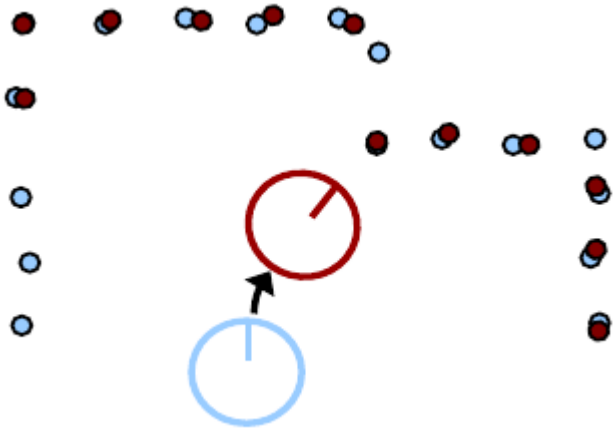
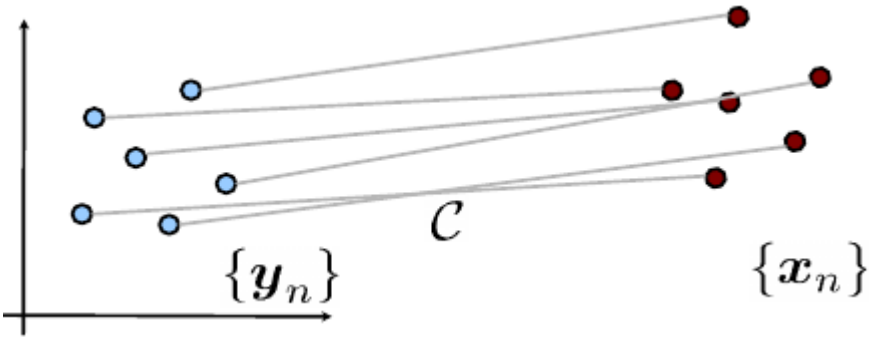
Ro07: LiDAR Odometry

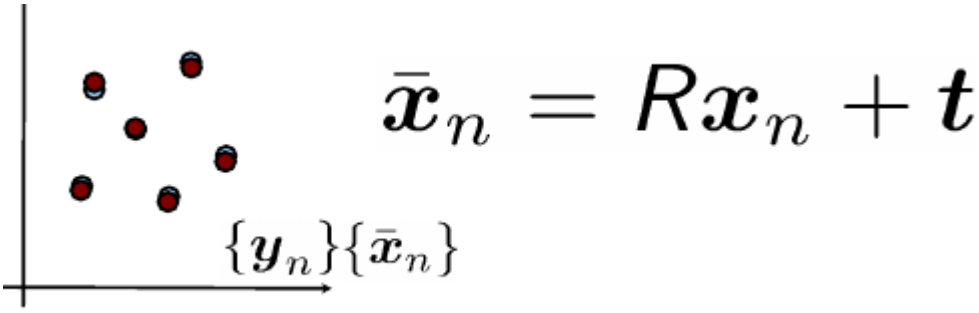
Pose Estimation from Scan Matching

- Consider robot at initialisation
- Observation of surroundings with range scanner (e.g., a LiDAR sensor)
  - 
    - Robot moves and repeats
- Each scan yields point cloud
- With knowledge on correspondence between these clouds: Alignment via Least Squares Minimisation
  - 

Simple Form of Point Cloud Registration



$$\hat{\boldsymbol{x}}_{\boldsymbol{n}} = R\boldsymbol{x}_n + \boldsymbol{t}$$
$$\sum ||\boldsymbol{y}_n - \hat{\boldsymbol{x}}_{\boldsymbol{n}}||^2 \rightarrow min$$

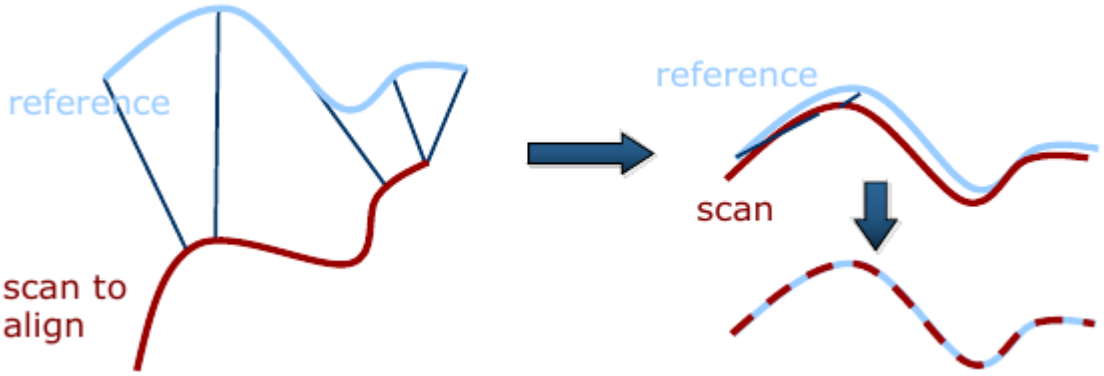


Tricky when Correspondence C unknown

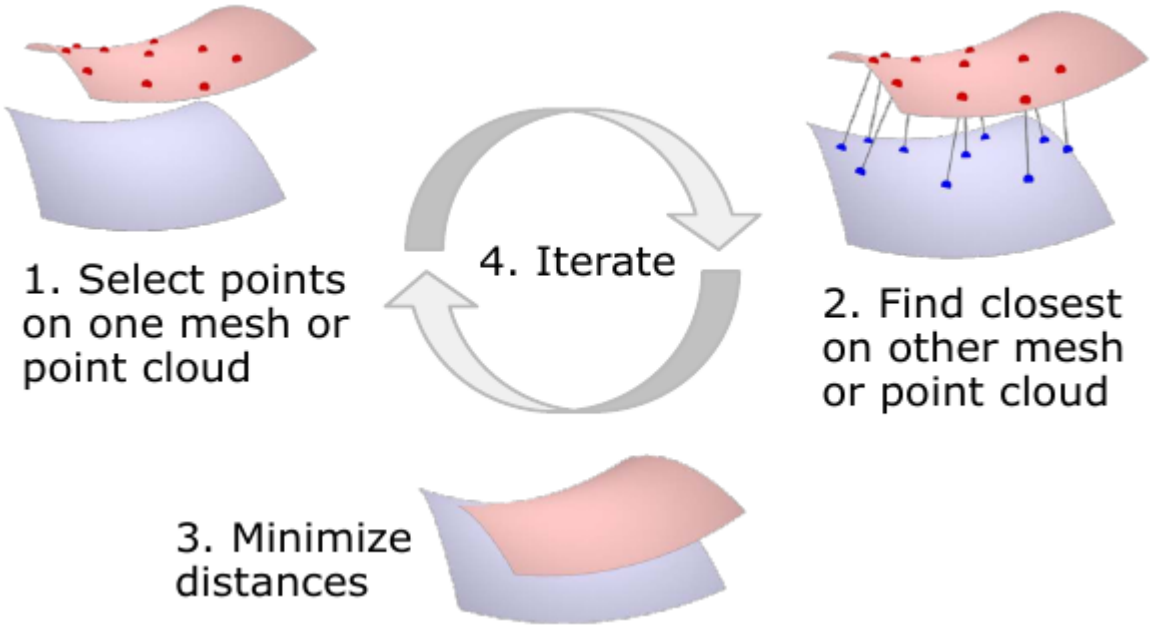
- Point Cloud Registration with **Known Data Association**
  - Derive efficient, optimal, direct solution:
    - Find parameters $\hat{\boldsymbol{x}}_{\boldsymbol{n}} = R\boldsymbol{x}_n + \boldsymbol{t}$
    - Squared error minimisation $\sum ||\boldsymbol{y}_n - \hat{\boldsymbol{x}}_{\boldsymbol{n}}||^2 p_n -> min$
  - Rotation $R = VU^T$
    - p_n are optional weights
  - Translation $\boldsymbol{t} = \boldsymbol{y}_0 - R\boldsymbol{x}_0$
    - with $H = \sum (\boldsymbol{x}_n - \boldsymbol{x}_0)(\boldsymbol{y}_n - \boldsymbol{y}_0)Tp_n$
      - $\boldsymbol{y}_0 = \frac{\sum \boldsymbol{y}_n p_n}{\sum p_n}$
      - $\boldsymbol{x}_0 = \frac{\sum \boldsymbol{x}_n p_n}{\sum p_n}$
    - $svd(H) = UDV^T$
    -
- Point Cloud Registration with **Unknown Data Associations**
  - No direct and optimal solution exists

Iterative Closest Point (ICP)

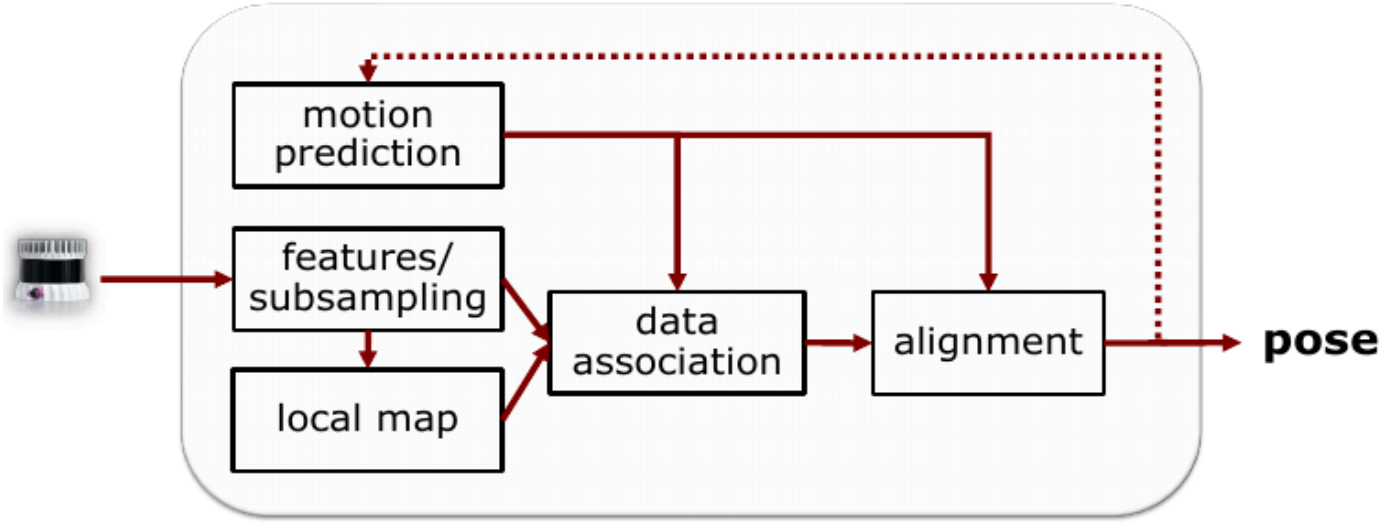
- If correct correspondance **unknown** -> impossible to determine **optimal parameters**
  - **Iteratively estimate data association and transformation given the data association**

- 

  1. Pick for every point its closest neighbor in the other point cloud ("closest point")
  2. Compute rigid body transform & align clouds
  3. Repeat
- If initial point clouds/correspondences are "close enough" -> Convergence



KISS - ICP - Keep It Small Simple

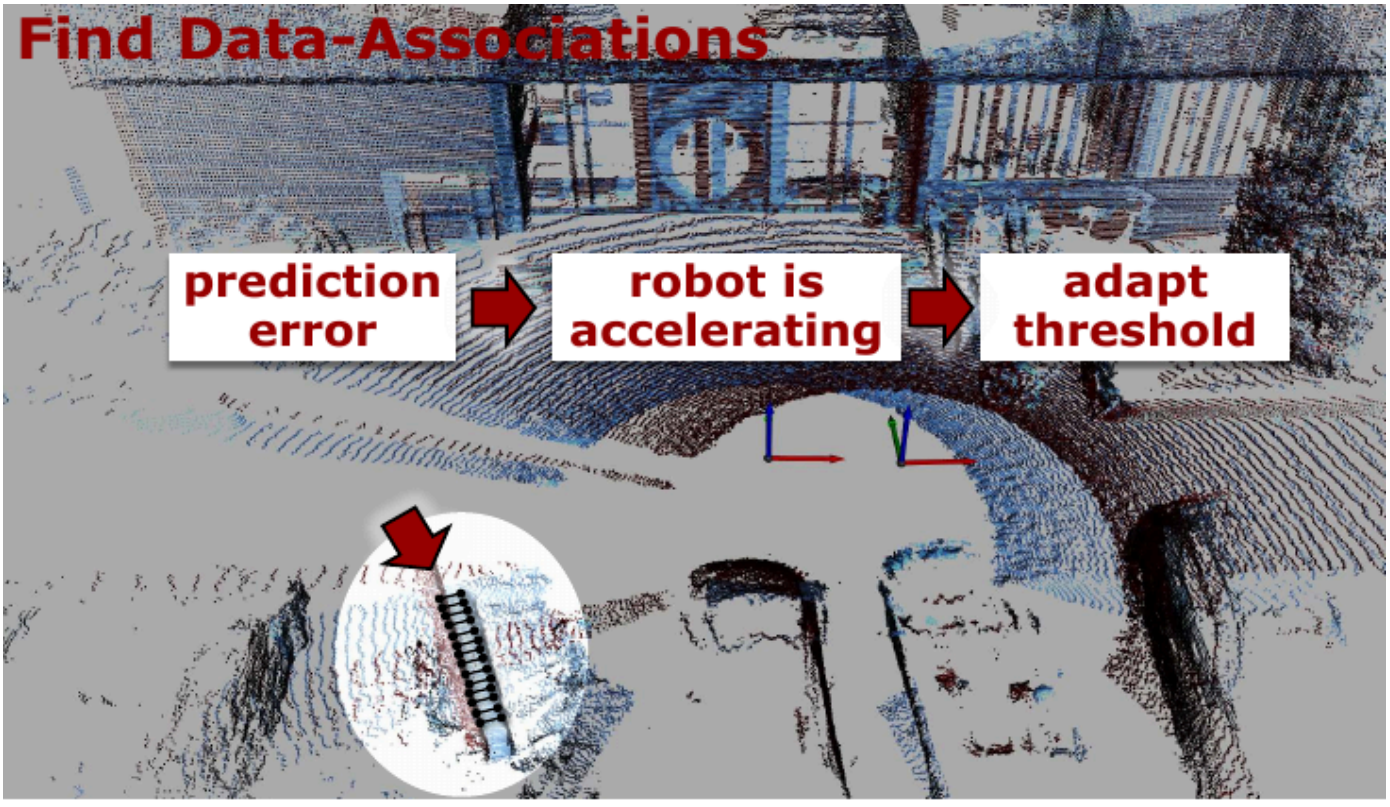


- Based on **point-to-point ICP** from 1991
  - ICP stripped down to minimal setup

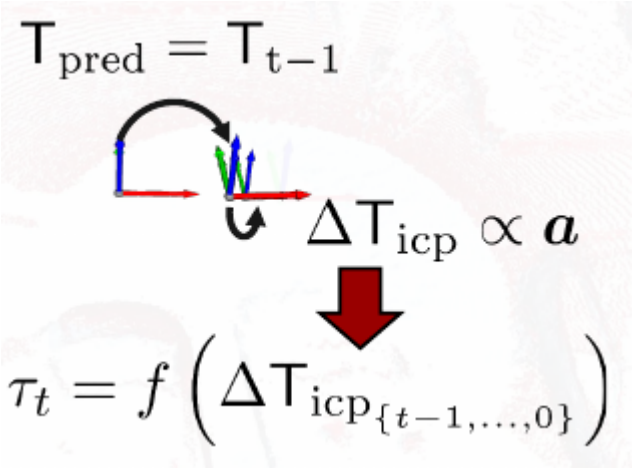
    - Only essential, but well done components
- **Generalises well** to different sensors
    - Works with all 3D LiDARs, RGBD, radar, ...
- **5 parameters** only

Key Ingredients of KISS-ICP

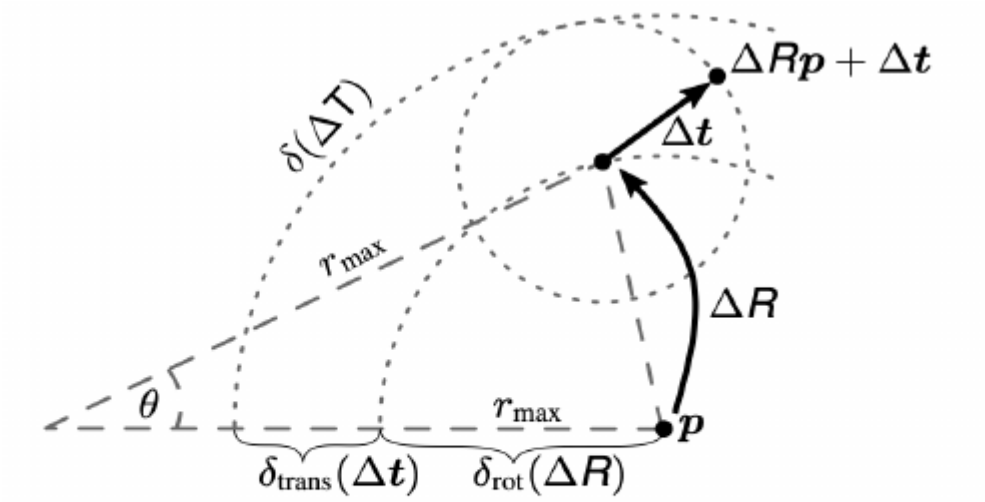
- **Motion prediction and scan de-skewing**
    - Constant velocity model de-skewing and prediction
    - Registration to correct deviation from the prediction
    - Possible extension: integrate IMU / VO / ...
- **Point cloud subsampling**
    - Voxel grid subsampling of each scan for registration
- **Local map and correspondence estimation**
    - Sliding voxel grid as local map
    - Register downsampled scan against local map
    - Maintian only N(=20) scans per voxel
    - Adaptive correspondence search based on statistics of registrations done so far
- **Alignment through robust optimisation**
    - Least squares with Gmean-McClure kernel
        - or even more fancy robust optimisation



Adaptive Threshold

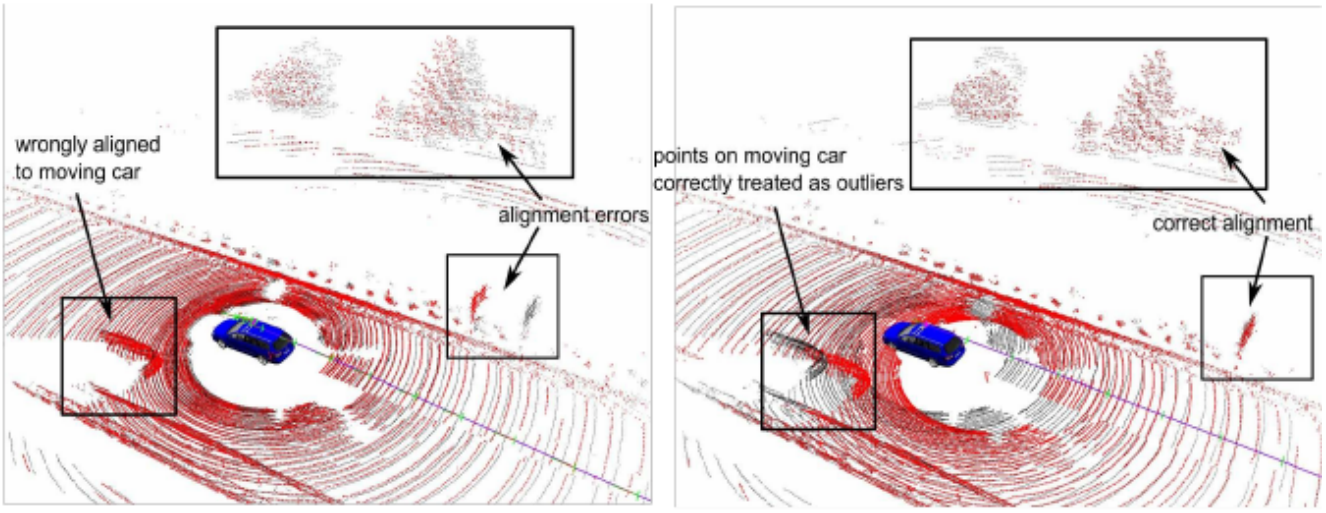


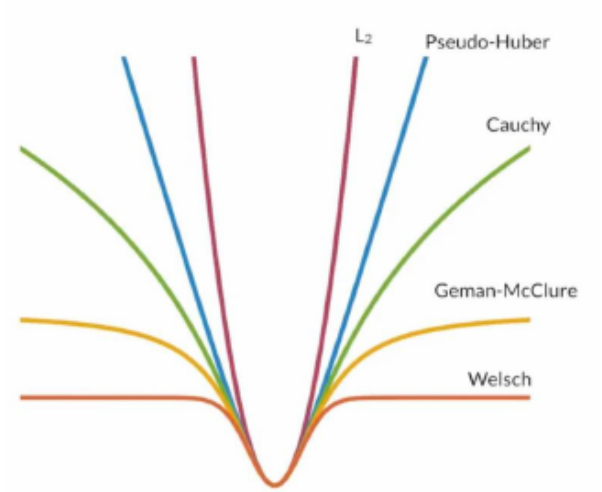
Search Area of Correspondences



$$\delta_{\text{rot}}(\Delta R) = 2\, r_{\max} \sin\left(\frac{1}{2}\underbrace{\arccos\left(\frac{\text{tr}(\Delta R) - 1}{2}\right)}_{\theta}\right) \qquad \delta_{\text{trans}}(\Delta\boldsymbol{t}) = \|\Delta\boldsymbol{t}\|_2$$

Outlier Handling with Robust Cost Kernels

- Outlier rejection in presence of dynamic objects
- 

- Relax least squares formulation
- Downweigh potential outliers
- 

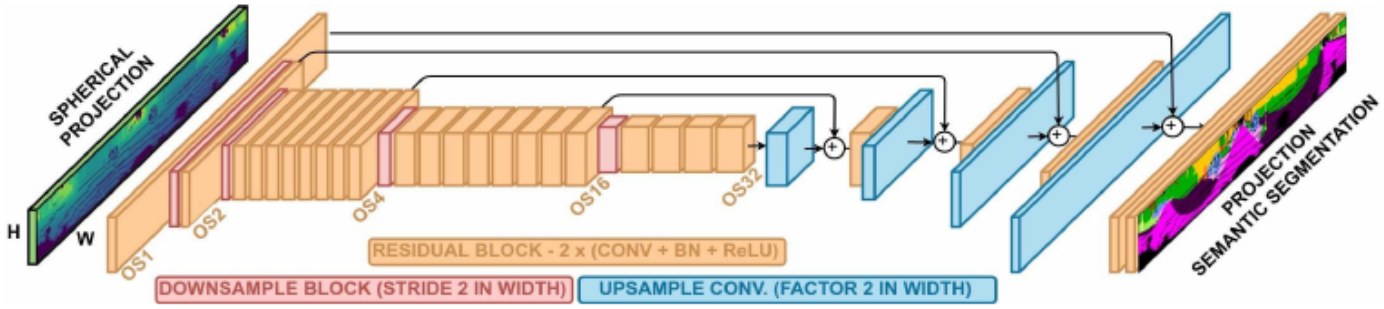
Which robust kernel to use?

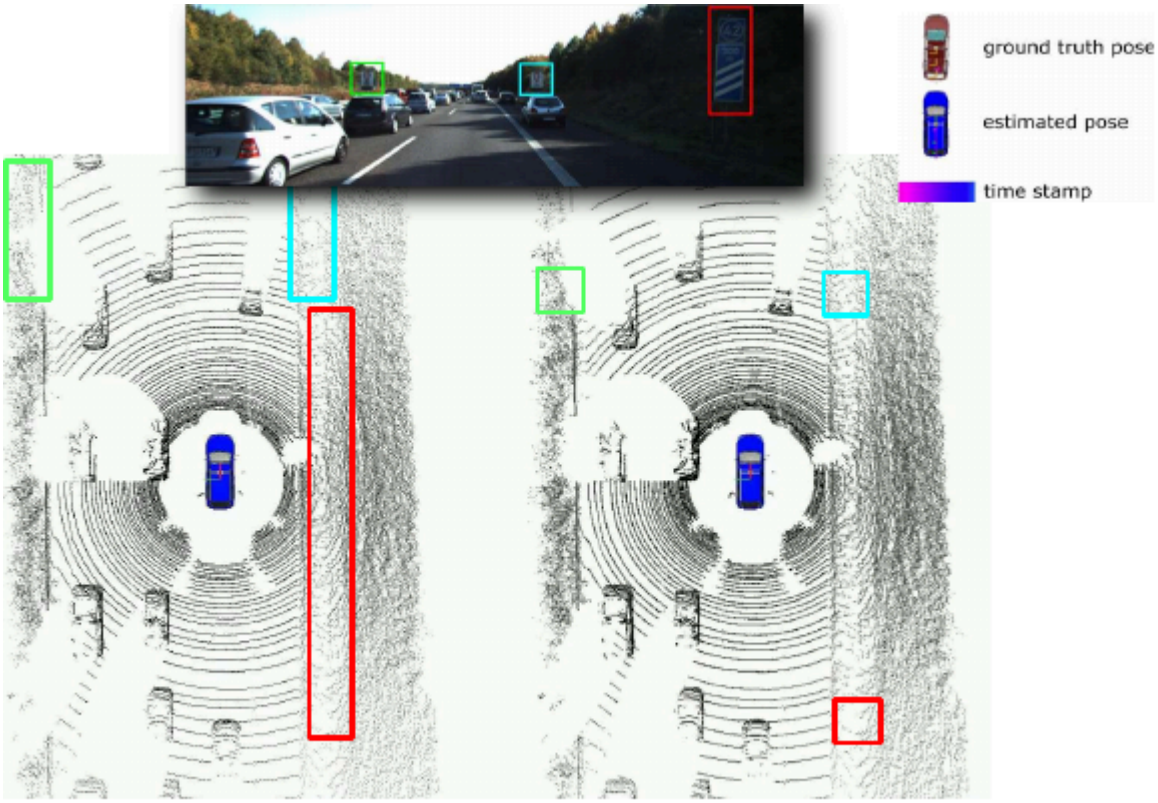
$$\rho(e,\alpha,c) = \frac{|\alpha-2|}{\alpha}\left(\left(\frac{(e/c)^2}{|\alpha-2|}+1\right)^{\alpha/2}-1\right)$$

Make the kernel shape part of the optimization problem

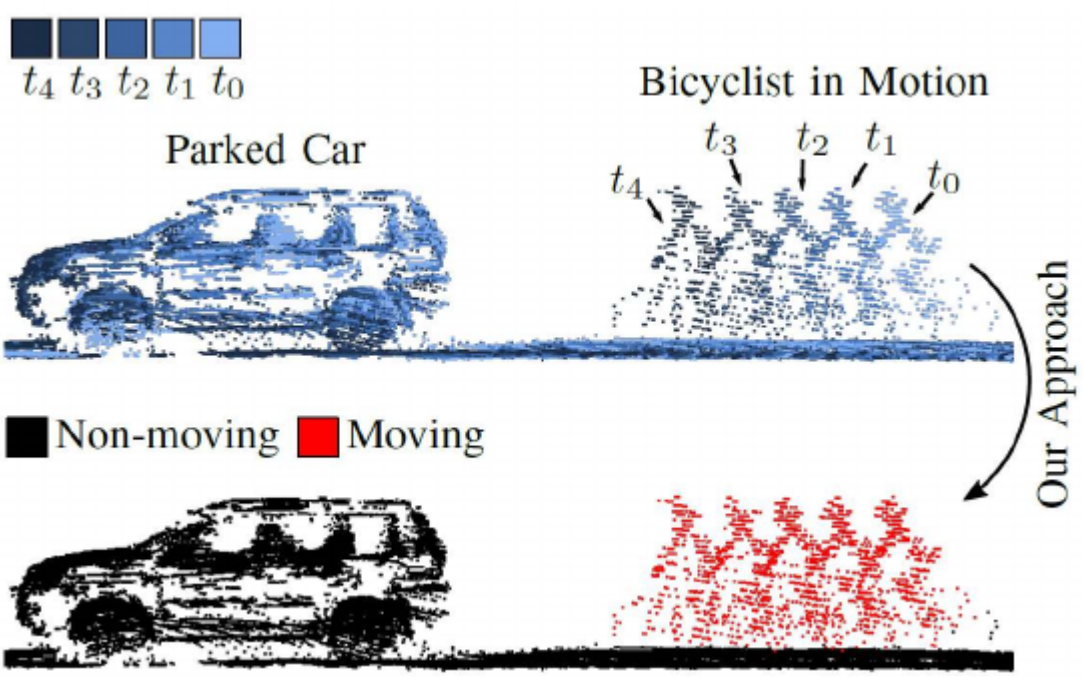
(Fast) Semantic Segmentation on Range Images

Projection-based





Geometry only + Semantics

- Idea: seperate moving from non-moving objects
- Need to look at sequential data
  - which objects are moving?
  - Ego-motion or object motion
- Register Scans:
  - 4D point cloud
  - Distinguish moving from non-moving points
- Moving Object Segmentation Using SParse 4DCNN
  - 

Summary

- Estimate which parts of scene are static and which are dynamic (heuristics, learning-based, ...)
- Mask point belonging to dynamic objects
- Perform an informed motion prediction
- Compute a good initial guess of the data association
- Run ICP with outlier rejection & robust optimisation